Bokinela Praneeth Kumar

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.57
Sri Chaitanya Junior kalasala	Class XII (Board of intermediate education, Andhra Pradesh, India)	2024	97.30%
Dr. K K R GOWTHAM SCHOOL	Class X (Board of secondary education, Andhra Pradesh, India)	2022	92.17%

PROJECTS

xv6 Kernel Extensions & Memory Management

- Extended the xv6 kernel with custom system calls, a System-Call-Aware Multi-Level Feedback Queue (SC-MLFQ) scheduler, and kernel statistics.
- Implemented virtual memory enhancements including page replacement, disk-backed swapping, FCFS/SSTF disk scheduling, and RAID-backed swap storage.
- **GitHub:** github.com/Praneeth740/memory-management

Transaction Management System (in an existing codebase)

- Implemented transaction support with BEGIN, COMMIT, and ABORT using an in-memory undo logging mechanism.
- Added rollback for INSERT, UPDATE, and DELETE operations while maintaining index consistency across multiple tables.
- **GitHub:** github.com/Praneeth740/transaction-management-

Accelerating C Programs using Idle Floating-Point Units

- Developed mixed integer-floating point implementations of computational kernels to improve utilization of idle FP units on a simulated RISC-V processor.
- Evaluated performance of software optimizations including loop unrolling and parallel accumulators using gem5 O3CPU performance metrics.
- **GitHub:** github.com/Praneeth740/idle-fpu-optimization

Arbitrary Precision Arithmetic Library

- Developed a Java library for arbitrary-precision integer and floating-point arithmetic beyond native data type limits.
- Implemented efficient algorithms for addition, subtraction, multiplication, and division on large numbers.
- **GitHub:** github.com/Praneeth740/arithmetic-java

SKILLS

- **Programming Languages:** C, C++
- **Database:** PostgreSQL
- **Web Technologies:** HTML, CSS
- **Version Control & Tools:** Git, GitHub, LaTeX

CODING AND ACHIEVEMENTS

- **LeetCode:** Solved 260+ problems. Profile
- **Codeforces:** Solved 220+ problems. Profile

RELEVANT COURSES

- **Computer Science:**Intro to Programming, Data Structures and Algorithms, Design and Analysis of Algorithms, Operating Systems, Database Management Systems, Theory of Computation, Computer Architecture
- **Mathematics:** Discrete Mathematics, Calculus, Linear Algebra, Probability and Random Variables, Introduction to Statistics

EXTRACURRICULAR

- Volunteer, National Service Scheme (NSS), IIT Hyderabad.
- Participated in marathon and community service activities.
- Participated in CPR and public health awareness programs.